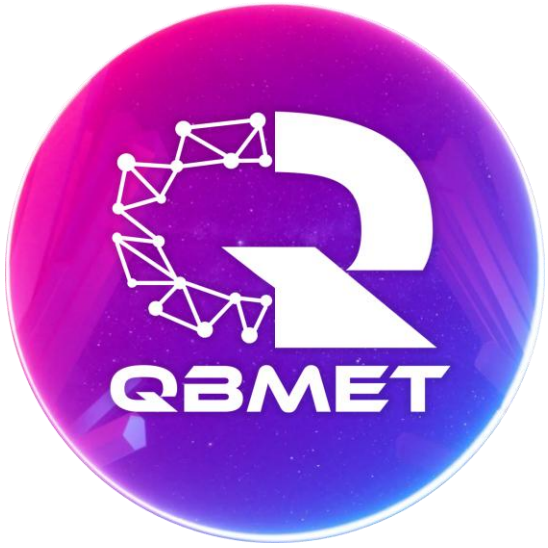




Application of CubeMars AK Series Robotic Actuators for a Lower Body Assistive Exoskeleton

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Introduction

Team Overview

At Queen's Biomechatronics team, we fuse mechatronics, robotics, and human-centered design to create technology that genuinely improves lives. Since 2021, our team of 70+ students has been working on developing active lower-limb exoskeletons to enhance strength and prevent occupational injuries in firefighters.

Current Design

To reduce the risk of musculoskeletal injury in first responders, we designed an active four-degrees-of-freedom lower-limb exoskeleton device. Constructed from lightweight carbon fiber and cost-effective 3D-printed joints, the exoskeleton frame anchors at the user's waist and houses AK60-6 V3.0 KV80 CubeMars brushless DC motors at each hip and knee joint. Adjustable waist circumference, thigh length, and shank length ensure anthropometrically aligned fit and improve device accessibility. Gait data is collected by MPU6050 3-axis inertial measurement units located below each hip and knee joint and is processed by an ESP32 microcontroller. Principal Component Analysis (PCA) is performed to characterize movement patterns in real time, and a Proportional-Derivative (PD) algorithm regulates the exoskeleton motor velocities to enhance natural limb movements based on measured angular velocities. The motors and microcontroller are powered by a 24V lithium-ion battery. To ensure user safety, an emergency stop button allows deactivation of the exoskeleton, while software limits and mechanical stops on each joint prevent movement beyond natural ranges of motion. Our current prototype was projected to produce a 10.1% efficiency increase for 45° knee flexion and a 14.4% efficiency increase for 45° hip flexion, while maintaining ranges of motion of 130° hip flexion, and 90° knee flexion to support human biomechanical principles. This prototype shows promise as a cost-effective exoskeleton for musculoskeletal support and injury prevention.

Problem Definition

During competition last year, QBMET used the AK60-6 V3.0 KV80 motors for both hips and knee joints on the suit. The team discovered issues using these motors on the hip joint. Firstly, due to the high load inertia created by movement of the leg on the hip joint, the low-torque motors had a minimal effect on the actuation of the hip. To find any effect, the hip joint motors were operated at their max rated torque (9Nm) for periods of up to 30 minutes. This created unforeseen issues where the motors would turn off due to overheating. The AK60-6 V3.0 KV80 worked well on the knee joints due to the lower torque command required and a lower load of inertia at the knee joint.

Design Proposal

To solve the issues the team experienced in our competition last year, QBMET proposes using higher torque motors on both the hip and knee joints. Specifically, the AK10-9 V2.0 KV60 motors on the hip joints, and the AK70-10 KV100 motors on the knee joints. How these motors were selected and how they will be implemented into the new exoskeleton design will be discussed in this report.

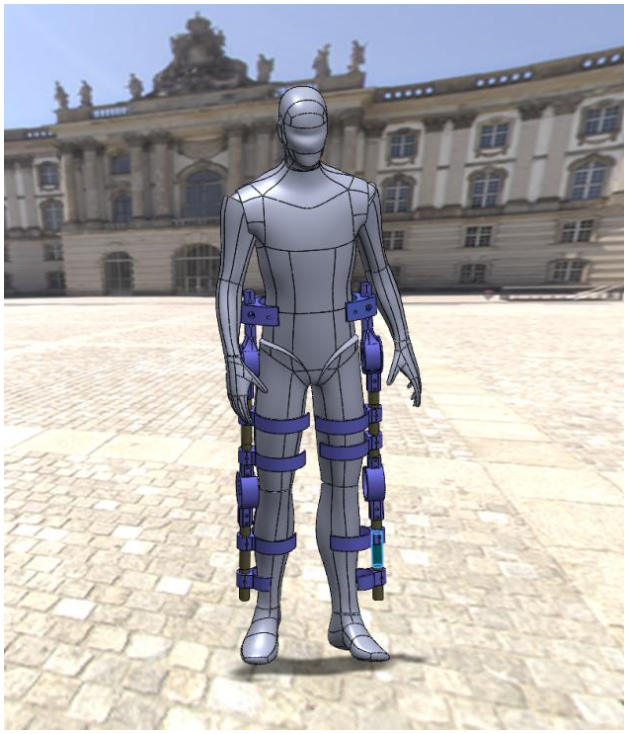


Figure 1: CAD Rendering of Current Design

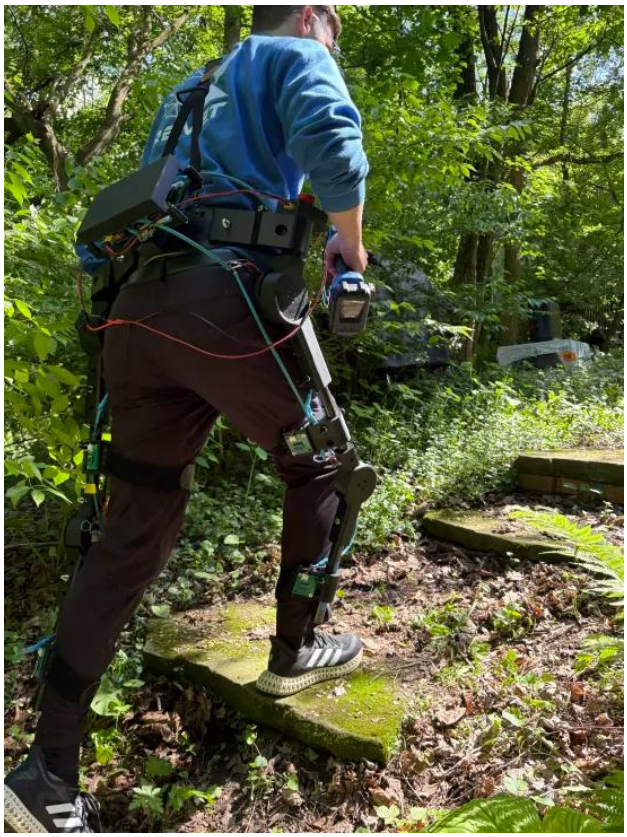


Figure 2: Photo of Current Design

Actuator Selection

Mechanical Considerations

During the design of each joint actuator, several factors were considered, including weight, torque output, and rotational velocity range. As the suit is designed to enhance human motion, it is important to ensure that it does not inhibit motion while no external torque is being applied. The simplest way to reduce this is to keep the actuator at each joint small, light, and close to the body to ensure that their mass does not affect the user's balance. The torque at each joint needed to be maximized while ensuring the rotational velocity of each actuator surpassed 120 RPM.

After considering these aspects, both the AK10-9 V2.0 KV60 and the AK70-10 KV100 actuators were selected. The AK10-9 V2.0 KV60 was chosen to be used as the hip actuator as the human hip moves at lower speeds and can handle more torque input when compared to the knees. In contrast, the AK70-10 KV100 provided the high rotational speeds required of a knee during high intensity movements while maintaining a torque of ~25Nm at its peak. This arrangement of RPM and peak torque proved to be the most optimal for our exoskeleton.

As seen in Figure 3, the AK70-10 KV100 maintains similar dimensions to the AK60-6 V3.0 KV80, which dimensions can be seen in Figure 4. This smaller size is ideal for a knee joint to avoid unnatural weight distribution, especially since the knee moves at such fast velocities during explosive movements such as running and jumping. In addition, the AK70-10 KV100 has twice the torque density of the AK60-6 V3.0 KV80, measuring at 47.6 Nm/kg. This ensures the motors mass is offset by the torque it inputs into the user's leg.

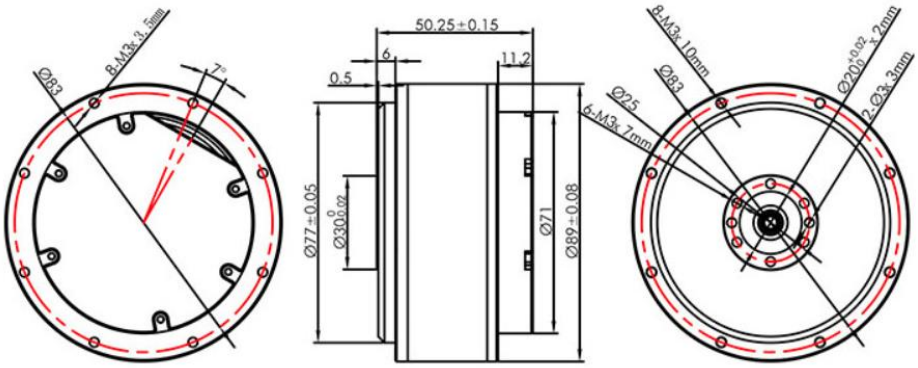


Figure 3: Dimensioned drawing of the AK70-10 KV100 taken from the CubeMars website.

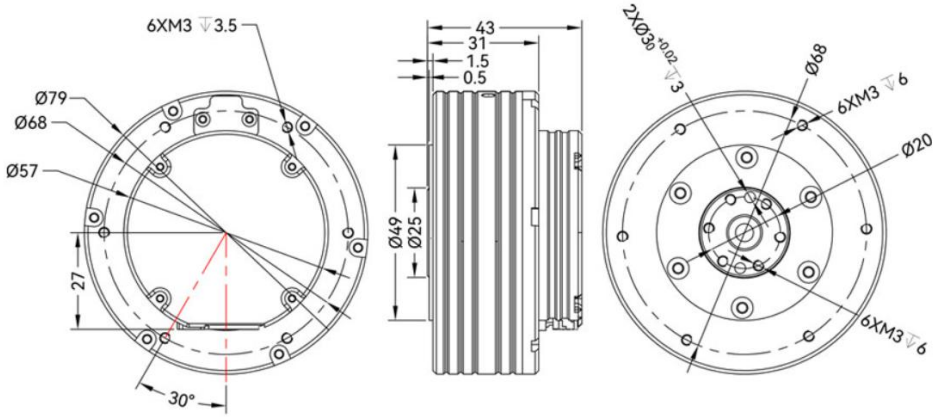


Figure 4: Dimensioned drawing of the AK60-6 V3.0 KV80 taken from the CubeMars website.

Although larger than the AK60-6 V3.0 KV80, the AK10-9 V2.0 KV60 provides over 5 times the peak torque, maintaining twice the torque density along with this. The additional mass is negligible since the hip motors maintain stationary relative to the user. In human kinetics, the hips are expected to generate significantly higher torque than the knees. Consequently, the AK10-9 V2.0 KV60 enables a more natural torque distribution, particularly in low-speed, high-force applications.

Electrical Considerations

As mentioned above, the suit currently uses four AK60-6 V3.0 KV80 motors, one at each hip and each knee. These are powered by a 24V, 4AH drill battery. We require the battery to last for approximately 30 minutes. With the implementation of these new motors, the electrical team plans to increase the battery capacity due to the increase in power required.

Using the ratio of the current motor and battery setup, we can extrapolate the expected battery capacity increase needed to maintain the required operating time. While the previous system experienced thermal issues within the hip motors due to high torque requirements, these issues were not related to the overall battery/power system implemented. The knee joint motors and batteries operated as expected and these will be used for a baseline comparison. These two motors have a max current draw of 6.7A.

Applying a conservative assumption of continuous operation at rated current to the original system yields a lower bound run time of approximately 18 minutes.

$$t = \frac{4Ah}{2(6.7A)} = 0.15h = 17.9min$$

During competition, the system exceeded this estimate and operated for the required 30 minutes, proving that this assumption underestimated real-world performance and provided a good outline for modelling the new system. Using this model, the proposed system has an operation time of 20 minutes, with real-world performance exceeding it.

The proposed configuration replaces the original four motors with two hip motors (AK10-9 V2.0 KV60) rated at 10.6A and two knee motors rated at 7.2A (AK70-10 KV100). To accommodate the increased current draw, the battery system will be upgraded from one 4Ah battery to two 24V, 6Ah batteries in parallel, resulting in a total capacity of 12Ah at the same operating

voltage as before. There is some supporting circuitry that is required when tying these batteries together and a design using series Schottky diodes is being investigated.

With the motors and batteries used last year, there were no electrical issues discovered. An analysis using the new motors and battery system was conducted assuming continuous operation at the rated current for each motor. This assumption assumes a worst-case scenario, as motors in the suit do not operate at the rated current continuously during most activities. Start up and shut down current spikes also contribute negligibly to the total battery life consumption.

While it is safe to assume that the motor is not constantly running at rated current, this approximation confirms that the more powerful motors will not adversely affect the battery life of the suit. Based on the conservative model of the new system and prior system performance, the upgraded battery system is expected to meet competition operation time requirements while supporting the higher current draw from the new motors.

The suit currently has "slow blow" safety fuses in line with each of the motors to stop overcurrent from flowing. The values of these fuses will be adjusted to match the new motors peak currents respectively.

Mounting and Integration

Mechanical

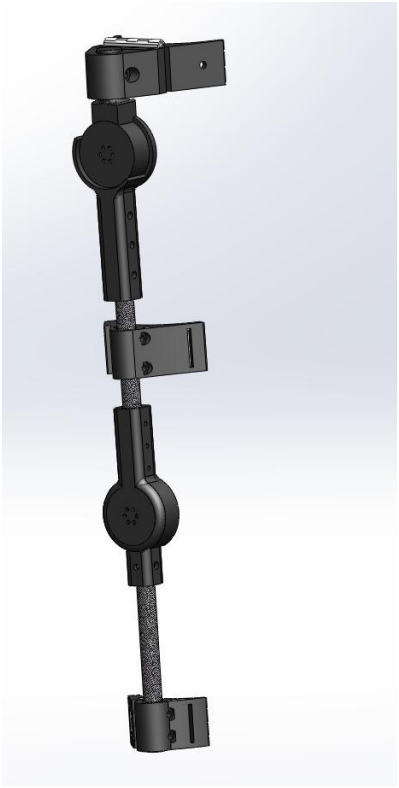




Figure 5: Image of single sided full exoskeleton assembly model. All CAD completed in SolidWorks.

Figure 6: Image of single sided full exoskeleton assembly on user.

As shown in Figure 5, the exoskeleton uses a motor on each hip and each knee. Each motor is held in its individual 3D printed housing, which is then attached to the carbon fibre tube frame. The 3D printed housings allow for efficient in-house manufacturing while ensuring the suit is light weight.

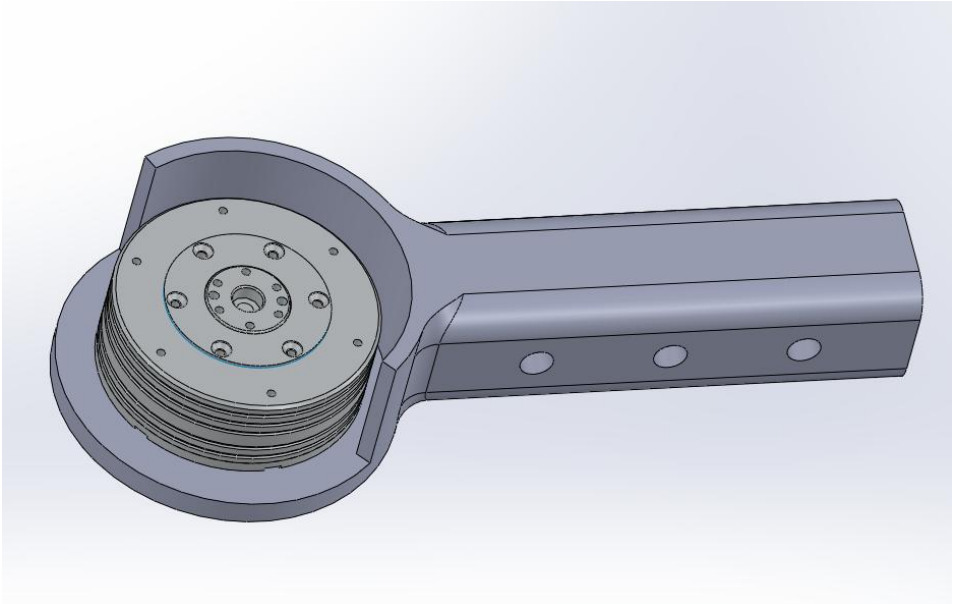


Figure 7: Image of hip motor housing with motor attached. All CAD completed in SolidWorks.

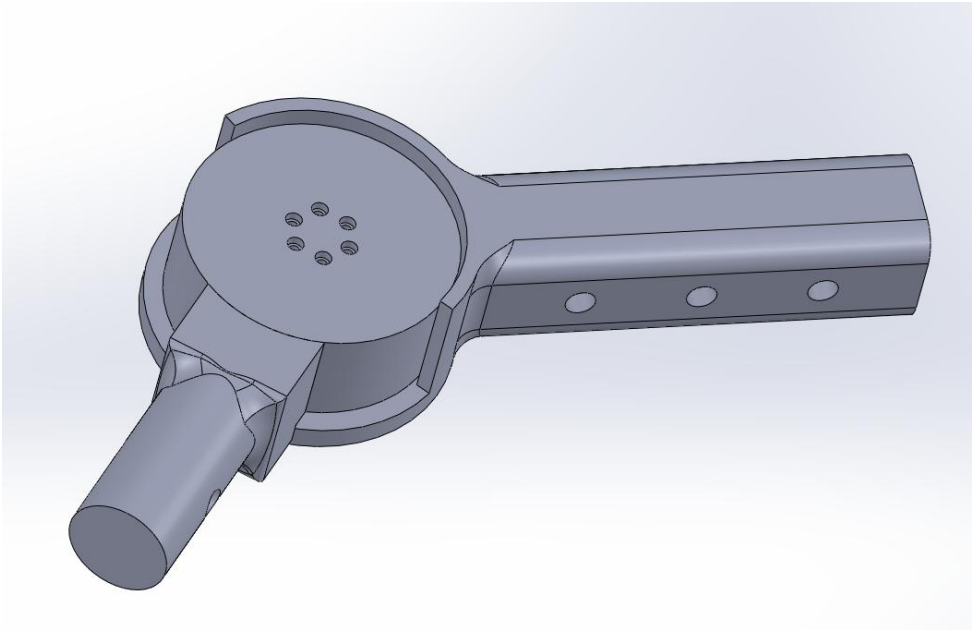


Figure 8: Image of fully assembled hip motor housing. All CAD completed in SolidWorks.

A detailed view of the motor housing can be seen in Figure 7, showing the opening allowing for hip and knee flexion. A second 3D printed housing is attached to the rotor and convers the motor, as seen in Figure 8. This design allows for each joint sub system to remain close to the body.

Electrical and Software

The current suit control system utilizes a cascaded admittance control architecture with each limb motor and gyroscope pair operating largely independently to the rest. The high-level

control loop, executed on the ESP32, processes filtered gyroscopic data to determine the user's limb velocity and applies a PD algorithm to compute a reactive torque. This torque is transmitted to the motor actuators via Controller Area Network (CAN) bus, feeding the low-level PD control system onboard the actuators. This configuration allows the motors to actively compensate for the suit's own impedance and providing additional assistance proportional to the user's movements while maintaining stability. The implementation of the proposed motors will not change as they operate under the same CAN system.

Electrical integration does not require modifications as it currently works with the lower power motors. The battery and fuse changes were described above and can be put in place as soon as needed.

Conclusion

Our team is requesting (2x) AK10-9 V2.0 KV60 and (2x) AK70-10 KV100 CubeMars actuators to use as the hip and knee actuators in our lower-limb exoskeleton. The increased torque in these motors is required to overcome the load inertia of the user's legs and provide the user with detectable movement support. The selected actuators will integrate into our existing system with minimal design changes while aligning with human biomechanics. This proposed motor upgrade will significantly enhance our existing design and ensure that our Queen's Biomechatronics Team maintains our competitive standing in international competitions.

Team Information

Executive Team Directory

Name	Title	Discipline
Amy Browne	Co-Captain	Engineering Physics
Jason Debbo	Co-Captain	Electrical Engineering
Will Hurley	Power Systems CTO	Electrical Engineering
Nathan Duncan	Software CTO	Mechatronics Engineering
Henry Bourdeau	Structural CTO	Mechanical Engineering
Eric McEvoy	Research Manager	Life Science
Vanessa Yeung	Outreach Manager	Bio-Mechanical Engineering
Mackenzie Conway	Marketing Manager	Bio-Mechanical Engineering
Callum McDougall	Marketing Manager	Film
Eric Filliol	Finance Manager	Electrical Engineering

Social Media Directory

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TikTok	@queens.biomechatr